



Pharmacologic Treatment of Obesity in Adults: Standards of Care in Overweight and Obesity

American Diabetes Association Professional Practice Committee for Obesity*

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Obesity medications may be part of a comprehensive care plan for adults with obesity. The Obesity Association, a division of the American Diabetes Association (ADA), developed comprehensive, evidence-based guidelines on the pharmacologic treatment of obesity in adults. When used in conjunction with lifestyle modifications, obesity medications have demonstrated efficacy in inducing and sustaining weight reduction while concurrently improving clinical outcomes of obesity and obesity-related diseases and complications. Health care professionals should engage people with obesity in a person-centered, shared decision-making approach when selecting an obesity medication to optimize health outcomes while emphasizing individual needs and preferences. The ADA's Obesity Association encourages health care professionals to adopt these guidelines for treatment of obesity in adults.

Obesity medications are an essential component of a comprehensive approach to obesity management, offering significant benefits beyond lifestyle interventions alone for many people. Research indicates that pharmacotherapy can lead to greater weight reduction and improved weight maintenance than placebo in adults with obesity (1). Of note, randomized controlled trials (RCTs) of obesity medications include lifestyle intervention in both placebo and study drug arms, which most typically comprises a 500-calorie deficit meal plan and behavioral intervention that usually produces 2.6% weight reduction in the placebo group (2), and the placebo group receives a placebo pill or injection, as appropriate. In meta-analyses of RCTs, people treated with obesity medications experienced substantial weight reduction compared with placebo (2,3). Participants

treated with obesity medications also had improvements in cardiometabolic markers, such as glycemia and lipid profiles (2,3), and some obesity medications demonstrated improvements in cardiovascular outcomes and other obesity-related diseases (4–7). When used appropriately, obesity medications offer a favorable risk-benefit profile in many cases, making them a viable option for many people with obesity.

Multiple weight-regulating hormones change in response to weight reduction, creating a physiologic environment conducive to the body returning to its higher prior weight (8,9). Weight reduction also negatively affects energy expenditure and basal metabolic rate, which makes weight-loss maintenance challenging for individuals with obesity (10,11). However, obesity medications enhance the ability to reduce weight and maintain weight reduction over extended periods (12), and many target the dysregulated neurohormonal systems that cause weight gain and prevent sustained weight reduction (13). Over the past few decades, substantial progress has been made regarding the efficacy and safety of obesity medications (14), and multiple obesity medications are now available that result in sustained weight reduction and improvements in multiple obesity-related diseases and complications. By discussing and considering obesity medications for adults with obesity, health care professionals can offer an effective treatment strategy that addresses overall health where treatment goals extend beyond weight reduction to include improving obesity-related diseases and complications, physical function, and well-being.

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Discussing Obesity Medications

Before prescribing an obesity medication, health care professionals should carefully review individuals' concomitant

TABLE 2.1 Summary of features of FDA-approved obesity medications influencing medication selection

Medication	Contraindications	Placebo-subtracted weight reduction in meta-analysis*	Effects on obesity-related diseases and complications†		Estimated cost‡§	Common adverse effects	Strategies to manage adverse effects§	Clinical suggestions and monitoring
			obesity-related diseases and complications†					
Orlistat	• Malabsorption • Cholestasis	3.1%	Demonstrated benefit • BP Potential benefit • T2D	Moderate	• Oily fecal spotting • Oily stools • Flatulence • Fecal urgency • Increased defecation	• Follow low-calorie meal plan with <30% of calories from fat to reduce GI effects • Avoid high-fat meal plan, as it increases adverse effects (e.g., ketogenic, low-carbohydrate)	• Advise that GI adverse effects typically subside within 4 weeks • Use MVI containing fat-soluble vitamins at bedtime to avoid deficiencies	
Phentermine	• CVD • Uncontrolled HTN • Glaucoma • Hyperthyroidism • SUD • Agitated states	3.6%	Unknown	Low	• Dry mouth • Insomnia • Headache • Irritability • Constipation	• Adequate water and dietary fiber intake to manage dry mouth and constipation • Avoid taking medication in the evening to reduce insomnia	Monitor • Heart rate and BP for increases with initiation and dose escalation	
Phentermine-topiramate	• CVD • Uncontrolled HTN • Glaucoma • Hyperthyroidism	8.8%	Potential benefit • T2D • BP • OSA	Moderate	• Paresthesia • Dry mouth • Headache • Constipation • Insomnia • Altered taste • Cognitive impairment	• Adequate water intake to manage dry mouth, constipation, and reduce risk of kidney stones • Adequate fiber intake to manage constipation • Take medication in the morning to reduce insomnia • Paresthesia or cognitive changes, if bothersome, may improve with reduced dose	• Advise people of childbearing potential of teratogenicity and counsel on need for effective contraception • Monitor • Heart rate and BP for increases with initiation and dose escalation • Mood changes • Electrolytes for changes in renal function	
Naltrexone-bupropion	• Uncontrolled HTN • Seizure disorders • Glaucoma • Anorexia or bulimia • SUD • Chronic opioid use	4.8%	Potential benefit • T2D No benefit • BP	Moderate	• Nausea • Constipation • Vomiting • Headache • Dizziness • Dry mouth	• Reduce portion size to manage nausea • Adequate water and dietary fiber intake to manage dry mouth and constipation • Administer evening dose at least 3 h before bedtime to reduce insomnia • Avoid high-fat meal plan, as it increases adverse effects (e.g., ketogenic, low-carbohydrate)	Monitor • BP for increases with initiation and dose escalation • Mood changes	
Liraglutide	• Severe GI disease • Personal or family history of medullary thyroid carcinoma or MEN type 2	4.5%	Demonstrated benefit • T2D Potential benefit • BP • ASCVD • MASH • OSA • OA	High	• Nausea • Vomiting • Diarrhea • Constipation • Headache • Dyspepsia • Increased heart rate	• Reduce portion size to manage nausea • Adequate water and dietary fiber intake to manage constipation • Last meal at least 2 h before bedtime to reduce dyspepsia	• Rotate injection site location to reduce pain • Refrigerate medication • Advise people on concurrent insulin or sulfonylureas of hypoglycemia risk • Monitor • Heart rate • Mood changes	

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TABLE 2.1 Summary of features of FDA-approved obesity medications influencing medication selection (Continued)

ASCVD, atherosclerotic cardiovascular disease; BP, blood pressure; CVD, cardiovascular disease; FDA, U.S. Food and Drug Administration; GI, gastrointestinal; HFpEF, heart failure with preserved ejection fraction; HTN, hypertension; MASH, metabolic dysfunction-associated steatohepatitis; MEN, multiple endocrine neoplasia syndrome; MVJ, multiple vitamin and mineral; OA, osteoarthritis; OSA, obstructive sleep apnea; SUD, substance use disorder; T2D, type 2 diabetes. *Values reflect placebo-subtracted mean percent weight loss estimated in meta-analyses to represent the weight reducing effect of the obesity medication beyond that achieved with lifestyle change alone (51/52). These meta-analyses include individuals with obesity both with and without obesity-related diseases and complications. Lifestyle counseling achieved 2.6% weight reduction in meta-analyses (2). Given that obesity medications should be used in combination with lifestyle changes, the total anticipated weight reduction effect would be the combined effect of the obesity medication plus lifestyle (e.g., 3.1% for orlistat plus 2.6% for lifestyle counseling would total 5.7% anticipated weight loss). Intensive behavioral therapy typically results in greater magnitude of weight reduction relative to lower intensity lifestyle counseling; therefore, the combination of obesity medication with intensive behavioral therapy may result in greater total anticipated weight reduction effect. Individual responses to obesity treatment may vary. †Obesity-related diseases and complications considered include T2D, BP, MACE, HFpEF, MASH, OSA, and OA. Obesity medications with A-level evidence of beneficial outcomes identified as “demonstrated benefit” for outcomes related to the specified condition, whereas those with B- or C-level evidence of beneficial outcomes identified as “potential benefit.” “No benefit” indicates that there is evidence of no beneficial effects on the outcome. If an outcome for one of the obesity-related diseases or complications is not listed, then the effect on this condition is unknown (i.e., there is no evidence evaluating the obesity medication on the outcome). ‡Individuals may experience cost barriers in accessing obesity medications due to a lack of insurance coverage, high copays, or other high out-of-pocket costs. Information presented in this column represents estimated out-of-pocket costs without insurance and using discount programs available from the manufacturer. “Low” is defined if estimated costs are less than \$100/month, “moderate” if \$100-\$200/month, or “high” if greater than \$200/month. Actual costs for the individual obtaining the medication may differ from these estimates. \$Most adverse effects will improve with reduction in obesity medication dose, and a slow dose escalation may reduce the likelihood of adverse effects and improve tolerability. Not all individuals need the maximum approved dose of the medication. ||These are medications currently FDA-approved for the treatment of the identified obesity-related disease or complication.

medications, as studies have found that medications that promote weight gain are commonly used, including among individuals with obesity (29–31). Health care professionals should be aware of weight-promoting medications (Table 2.2) (32). As the number of prescribed weight-promoting medications increases, individuals' BMI and waist circumference are more likely to have greater increases over time (33). In addition, studies have found that participants using weight-promoting medications have reduced weight-loss efficacy with lifestyle interventions to treat obesity (34,35).

Whenever possible and clinically appropriate, avoid initiating a prescription for weight-promoting medications among individuals with obesity and instead select a non-weight-promoting alternative. For example, ACE inhibitors, angiotensin receptor blockers, calcium channel blockers, and diuretics should be prioritized to treat hypertension rather than some β -blockers, as these options are weight-neutral rather than weight-promoting. When diabetes medications are needed for glycemic management, medications most likely to cause weight gain should be discouraged in favor of alternatives that are non-weight-promoting (Table 2.2). However, health care professionals should not refrain from prescribing weight-promoting medications when needed to achieve treatment goals for that condition (e.g., blood pressure in hypertension).

For individuals with obesity who may already be using weight-promoting medications, health care professionals should determine whether a dose reduction or non-weight-promoting alternative may be clinically appropriate. If so, individuals should be engaged in determining whether their medication plan should be modified, accounting for the benefits and risks of all treatment options. Health care professionals should be aware that cessation of weight-promoting medications does not necessarily result in weight reduction. For individuals who must remain on weight-promoting medications, obesity medications may be beneficial in achieving treatment and weight goals.

Setting Treatment Goals With Obesity Medications

Recommendations

2.5a Treatment goals should be incremental and individualized, based on the severity of obesity, obesity-related diseases and complications, and individuals' needs, life circumstances, and preferences **E**

2.5b Aim for a sustained reduction in baseline body weight of $\geq 5\%$ for adults treated with obesity medications, which may achieve some clinically meaningful health benefits **A**. Aim for a sustained weight reduction of $\geq 10\%$ to manage many obesity-related diseases or complications **A**; in some cases, a $\geq 15\%$ weight reduction may be indicated to achieve greater therapeutic benefit **B**.

Obesity is a heterogeneous disease with varying effects on health, necessitating individualized treatment plans.

Sociocultural factors and personal motivation also play a critical role in treatment goals. Culturally tailored interventions that respect individual preferences and lived experiences, rather than those that impose rigid weight-loss goals, are needed. Health care professionals should evaluate each individual for obesity-related diseases and complications as well as physical symptoms, psychological symptoms, functional limitations, and impact on well-being due to excess adiposity (36,37). Such a holistic assessment will highlight the need for tailored treatment goals, as health risks and weight-reduction benefits vary by the severity of obesity and its related diseases and complications. Health care professionals should be aware that the typical time course to achieve treatment goals varies between individuals; therefore, reevaluation every 3 months may be useful to monitor progress.

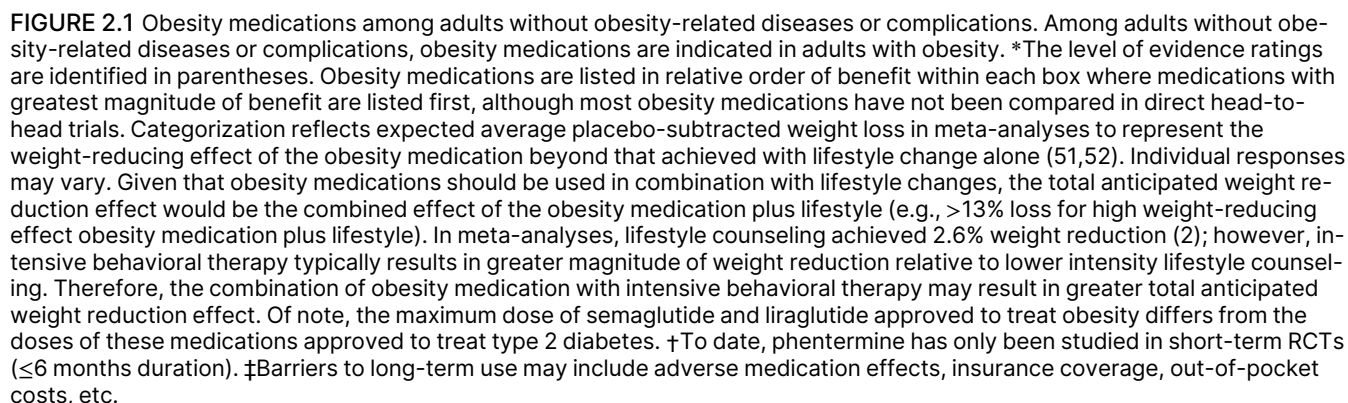
Achieving a $\geq 5\%$ reduction in baseline body weight for adults treated with obesity medications is associated with health benefits (18,38–42). Research demonstrates that a 5% weight reduction improves cardiometabolic risk factors (2,43,44). For example, the CONQUER RCT demonstrated that over 60% of participants treated with phentermine-topiramate achieved $\geq 5\%$ weight reduction as compared with 21% with placebo, and weight reduction was accompanied by improvements in metabolic risk factors (40). In the SURMOUNT-1 RCT, over 85% of participants who received tirzepatide achieved a $\geq 5\%$ reduction goal compared with 35% with placebo; notable improvements in cardiometabolic measures were also observed (18).

For individuals with obesity-related diseases and complications, achieving $\geq 10\%$ reduction in baseline body weight has consistently demonstrated improvements in these conditions. In general, 10–15% weight reduction results in significantly greater improvements in glycemia, blood pressure, and lipids compared with 5–10% loss (45). In most cases, the magnitude of weight reduction is directly proportional to risk factor improvements; therefore, sustained weight reduction of $\geq 10\%$ is often required for optimal cardiometabolic benefits. Obesity-related diseases and conditions that benefit from this weight-loss magnitude include metabolic dysfunction-associated steatohepatitis with moderate-to-severe fibrosis (7,46), heart failure with preserved ejection fraction (47,48), and knee osteoarthritis (49).

Emerging data suggest that $\geq 15\%$ weight reduction may be needed to achieve clinical benefits in some cases (45), particularly for individuals with moderate-to-severe obesity-related conditions. In the SURMOUNT-OSA trial, participants with obesity and moderate-to-severe obstructive sleep apnea who received tirzepatide achieved mean weight reduction exceeding 15% and experienced significant reductions in the apnea-hypopnea index (AHI), hypoxic burden, and systolic blood pressure compared with placebo (6). A meta-analysis found a dose-response relationship between AHI and weight

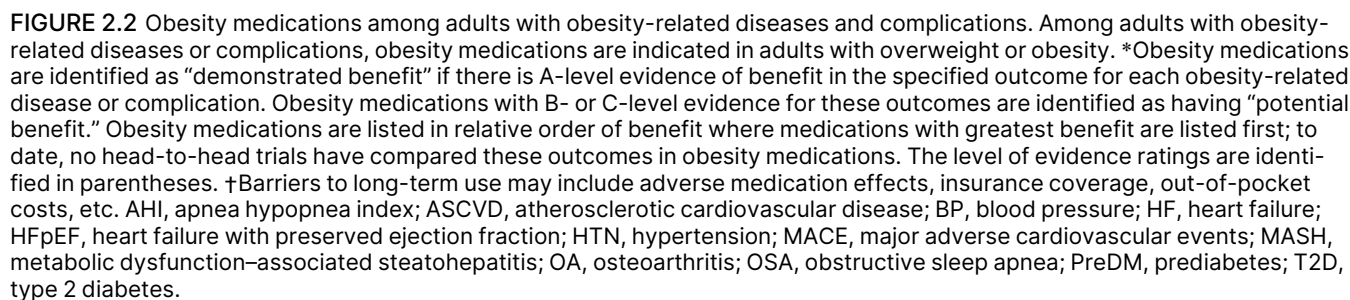
Condition	Weight-promoting medications	Non-weight-promoting alternatives
Hypertension	α -Blockers Metoprolol Propranolol Atenolol	ACE inhibitors ARBs Carvedilol Nebivolol Thiazide and thiazide-like diuretics
Diabetes	Insulin Sulfonylureas Thiazolidinediones	α -Glucosidase inhibitors Metformin* GLP-1 receptor agonists* Dual GIP and GLP-1 receptor agonist* SGLT2 inhibitors* Vidagliptin* Sitagliptin* Other DPP-4 inhibitors Pramlintide*
Contraception	Depo-medroxyprogesterone acetate Progestins	Copper IUDs Barrier contraceptive methods
Allergies	Diphenhydramine Cetirizine Hydroxyzine Fexofenadine Meclizine Cyproheptadine	Loratadine
Depression	Nortriptyline Doxepin Amitriptyline Imipramine Phenelzine Paroxetine Escitalopram Citalopram Fluoxetine† Sertraline† Mirtazapine	Bupropion*
Psychosis	Clozapine Olanzapine Risperidone Quetiapine Perphenazine Aripiprazole	Ziprasidone
Mood instability	Lithium	Ziprasidone Lamotrigine
Epilepsy	Gabapentin Pregabalin Valproic acid Vigabatrin Carbamazepine	Lamotrigine Levetiracetam Phenytoin Topiramate* Zonisamide* Felbamate*
Inflammation	Glucocorticoids	NSAIDs
Other	Megestrol acetate	

and personal circumstances. While weight-reduction goals of $\geq 15\%$ may be beneficial for some individuals with moderate-to-severe obesity-related diseases, others may experience meaningful health improvements with less weight reduction. Personalized treatment strategies may lead to long-term success and sustainability.



phentermine dose is stable, with follow-up every 3–6 months for long-term treatment to maintain health goals. Phentermine is one of the most prescribed obesity medications in the U.S. (58).

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decreasing intake of high-fat foods, increasing intake of high-fiber foods, avoiding long intervals between meals, and eating the last meal ≥ 2 h before bedtime (63,64).

For an individual with obesity who does not have any obesity-related diseases or complications, the primary treatment goal is to prevent the development of these

TABLE 2.3 Mechanisms of action and dosing of obesity medications for adults

Medication (route)	Mechanisms of action	Available doses	Manufacturer-recommended dose-escalation schedule*	Maximum approved dose	Dose adjustment in renal impairment†	Impact on oral contraceptive efficacy
Orlistat (oral)	Intestinal lipase inhibitor	120 mg 60 mg (OTC)	None	120 mg TID	Not required	None
Phentermine (oral)	Sympathomimetic amine	8 mg 15 mg 30 mg 37.5 mg	None	8 mg TID 37.5 mg daily	eGFR 15–29 mL/min: 15 mg daily max	None
Phentermine-topiramate (oral)	Sympathomimetic amine combined with GABA augmentation	3.75/23 mg 7.5/46 mg 11.25/69 mg 15/92 mg	Week 1–2: 3.75/23 mg daily Week 3–13: 7.5/46 mg daily Week 14–15: 11.25/69 mg daily‡ Week 16+: 15/92 mg daily	15/92 mg daily	CrCl <50: 7.5/46 mg daily max	Altered bioavailability to OCP components that may cause irregular bleeding
Naltrexone-bupropion (oral)	POMC neuron stimulation via dopamine/norepinephrine reuptake inhibitor and opioid receptor antagonist	8/90 mg	Week 1: 1 tablet daily Week 2: 1 tablet BID Week 3: 2 tablets QAM and 1 tablet QPM Week 4+: 2 tablets BID	16/180 mg BID	Moderate to severe renal impairment: 8/90 mg twice daily max	None
Liraglutide (SQ)	GLP-1 receptor agonist	18 mg/3 mL pen	Week 1: 0.6 mg SQ daily Week 2: 1.2 mg SQ daily Week 3: 1.8 mg SQ daily Week 4: 2.4 mg SQ daily Week 5+: 3.0 mg SQ daily	3.0 mg daily	Not required	None
Semaglutide (SQ)	GLP-1 receptor agonist	0.25 mg/0.5 mL pen 0.5 mg/0.5 mL pen 1 mg/0.5 mL pen 1.7 mg/0.75 mL pen 2.4 mg/0.75 mL pen	Week 1–4: 0.25 mg SQ weekly Week 5–8: 0.5 mg SQ weekly Week 9–12: 1 mg SQ weekly Week 13–16: 1.7 mg SQ weekly§ Week 17+: 2.4 mg SQ weekly	2.4 mg weekly	Not required	None
Tirzepatide (SQ)	Dual GIP and GLP-1 receptor agonist	2.5 mg/0.5 mL pen or vial 5 mg/0.5 mL pen or vial 7.5 mg/0.5 mL pen or vial 10 mg/0.5 mL pen or vial 12.5 mg/0.5 mL pen or vial 15 mg/0.5 mL pen or vial	Week 1–4: 2.5 mg SQ weekly Week 5–8: 5 mg SQ weekly§ Week 9–12: 7.5 mg SQ weekly Week 13–16: 10 mg SQ weekly§ Week 17–20: 12.5 mg SQ weekly Week 21+: 15 mg SQ weekly	15 mg weekly	Not required	Significant interaction with OCPs¶

BID, twice a day; CrCl, creatinine clearance; eGFR, estimate glomerular filtration rate; GABA, γ -aminobutyric acid; GIP, glucose-dependent insulintropic polypeptide; GLP-1, glucagon-like peptide 1; OCP, oral contraceptive pill; OTC, over the counter; POMC, proopiomelanocortin; QAM, every morning; QPM, every evening; SQ, subcutaneous; TID, three times a day. *In clinical practice, dose escalation or de-escalation should be tailored to individual treatment response balancing benefits and adverse effects. Information provided is based on manufacturer-suggested dose-escalation schedule; dose-escalation schedule and maintenance dose is often modified based on individual response and goals. †Information is adapted from manufacturer-suggested renal dose adjustment. For individuals with severe renal impairment (eGFR <30 mL/min) or end-stage kidney disease, caution and close monitoring should occur if orlistat, liraglutide, semaglutide, or tirzepatide is used. Phentermine, phentermine-topiramate, and naltrexone-bupropion should be avoided in these individuals. ‡For individuals who achieve $\geq 3\%$ weight reduction at week 14, phentermine-topiramate may be continued at 7.5/46 mg daily dose. §Indicated doses may also be maintenance doses per manufacturer guidelines. ||No effect was found on bioavailability of OCP; however, concern has been raised about how delayed gastric emptying from the medication may affect efficacy of OCPs. ¶Advise individuals using oral hormonal contraceptives to switch to a nonoral contraceptive method, or add a barrier method of contraception, for 4 weeks after initiation with tirzepatide and for 4 weeks after each dose escalation.

Medication	Maximum approved dose	Median AWP for 30-day supply† (range)	Median NADAC for 30-day supply† (range)
Orlistat‡	120 mg TID	\$651 (\$520–\$781)	\$518 (\$416–\$619)
Phentermine	37.5 mg daily	\$43 (\$3–\$56)	\$2 (\$2–\$3)
Phentermine-topiramate	15/92 mg daily	\$251	NA
Naltrexone-bupropion	16/180 mg BID	\$750	NA
Liraglutide	3 mg daily	\$1,619	\$1,304
Semaglutide	2.4 mg weekly	\$1,619	\$1,304
Tirzepatide	15 mg weekly	\$1,304§	\$1,052§

be noted that tirzepatide, semaglutide, liraglutide, and phentermine-topiramate, on average, cause small increases in resting heart rate (~2–4 bpm), although some individuals may experience larger increases (~10–20 bpm). This heart rate change is not clinically detrimental to most individuals, as the rate-pressure product (heart rate $\times$ systolic blood pressure) (96–98), which indicates workload on the heart, remains favorable due to blood pressure reductions. Health care professionals should be aware that naltrexone-bupropion does not reduce blood pressure commensurate with weight reduction (89), and both naltrexone-bupropion and phentermine are contraindicated in individuals with uncontrolled hypertension.

Obesity contributes to atherosclerotic cardiovascular disease (ASCVD) development through multiple mechanisms (99), and the duration of obesity and visceral adiposity are strong predictors of coronary artery disease in epidemiologic studies (100). The magnitude of weight reduction achieved may be particularly critical for ASCVD (99), as modest weight reduction with intensive lifestyle intervention did not reduce the rate of cardiovascular events in adults with T2D and obesity (101); a post hoc analysis of this RCT found that only participants who achieved $\geq 10\%$ weight reduction had significant reductions in cardiovascular events (102). In adults with obesity and established ASCVD, the treatment plan should include a GLP-1RA with demonstrated benefits or dual GIP/GLP-1RA with potential benefits in reducing cardiovascular events as well as weight reduction (Fig. 2.2 and Table 2.5).

established cardiovascular disease without diabetes (4). Further, semaglutide produced durable weight reduction (104) and significantly reduced new-onset diabetes and a composite kidney outcome (driven by reductions in new-onset macroalbuminuria) (105). While a cardiovascular outcomes trial evaluating liraglutide 3.0 mg has not been performed, liraglutide 1.8 mg (T2D-approved dose) reduces cardiovascular events in people with T2D at increased cardiovascular risk, including among participants with obesity (5). Similar findings were observed with liraglutide 3.0 mg versus pooled comparators in the SCALE trials (HR 0.42; 95% CI 0.17–1.08) (106), but this analysis was underpowered due to low event rates.

A cardiovascular outcomes trial of dual GIP/GLP-1RA obesity medication has not yet been published; however, there is evidence to support potential ASCVD benefits. Tirzepatide has been shown to significantly improve cardiovascular risk factors (e.g., blood pressure, cholesterol, systemic inflammation, and glycemic management) in people with obesity (18,107). In a SURMOUNT-1 trial post hoc analysis, tirzepatide significantly reduced the predicted 10-year risk of ASCVD events compared with placebo (108). Moreover, in a prespecified participant-level pooled analysis of the SURPASS program, tirzepatide did not increase the risk of major adverse cardiovascular events in people with T2D versus pooled comparators (HR 0.80; 95% CI 0.57–1.11) (109).

The LIGHT trial, which randomized people with BMI 27–50 kg/m² at increased cardiovascular risk to naltrexone-bupropion, was designed to evaluate cardiovascular outcomes. In the primary analysis (conducted after 50% of planned events), naltrexone-bupropion did not reduce major adverse cardiovascular events compared with placebo (110). However, due to unplanned early termination after public release of confidential interim data, the LIGHT trial was unable to rigorously establish the safety and efficacy of naltrexone-bupropion compared with placebo. Moreover, most participants permanently

TABLE 2.5 Summary of key evidence to inform obesity medication selection, by populations with specific obesity-related diseases and complications*

Obesity medication	Mean weight change with obesity medication* (time point)	Select outcomes relevant to population
Adults with obesity and prediabetes: progression to type 2 diabetes (Recommendation 2.7)		
Tirzepatide	−12.3% to −19.7% (176 weeks)	• 1.3% TZP vs. 13.3% PBO received a diagnosis of T2D at 176 weeks (HR 0.07; 95% CI 0.0–0.1; <i>P</i> <0.001) (73)
Semaglutide	−13.9% (52 weeks)	• 1% SEMA vs. 3% PBO progressed to T2D at 52 weeks (71) • 1.5% SEMA vs. 6.9% PBO progressed to T2D at 156 weeks (<i>P</i> <0.0001) (72)
Phentermine-topiramate	−10.9% to −12.1% (108 weeks)	• Annualized incidence rates for progression to T2D: 1.3% in PHEN-TOP 15 mg/92 mg, 1.8% in PHEN-TOP 7.5 mg/46 mg, and 6.1% in PBO at 108 weeks (66)
Liraglutide	−6.1% (160 weeks)	• 2% LIRA vs. 6% PBO were diagnosed with T2D at 160 weeks (HR 0.21; 95% CI 0.13–0.34; <i>P</i> <0.0001) (70)
Orlistat	−5.3% (208 weeks)	• Cumulative incidence of T2D was 6.2% ORL vs. 9.0% PBO at 208 weeks (HR 0.63; 95% CI 0.46–0.86) (69)
Adults with obesity and type 2 diabetes: glucose-lowering efficacy (Recommendation 2.8)		
Tirzepatide	−7.9% to −11.0% (40 weeks) −12.8% to −14.7% (72 weeks)	Mean A1C change • −1.9% to −2.1% TZP vs. +0.0% PBO at 40 weeks (84) • −2.1% TZP vs. −0.5% PBO at 72 weeks (78) Achieved A1C <7% • 87% to 92% TZP vs. 19% with PBO at 40 weeks (84) • Over 80% TZP vs. 36% PBO at 72 weeks (78)
Semaglutide	−4.2% to −4.7% (30 weeks) −7.0% to −9.6% (68 weeks)	Mean A1C change • −1.5% to −1.6% SEMA vs. −0.0% PBO at 30 weeks (85) • −1.5% to −1.6% SEMA vs. −0.4% PBO at 68 weeks (79) Achieved A1C <7% • 72% to 74% SEMA vs. 25% PBO at 40 weeks (85) • 72.3% to 78.5% SEMA vs. 26.5% PBO at 68 weeks (79)
Liraglutide	−4.7% to −6.0% (56 weeks)	Mean A1C change • −0.7% to −1.0% LIRA vs. +0.1% PBO at 26 weeks (254) • −1.1% to −1.3% LIRA −0.3% PBO at 56 weeks (<i>P</i> < 0.001) (80) Achieved A1C <7% • 66.7% to 69.2% LIRA vs. 27.2% PBO at 56 weeks (<i>P</i> < 0.001) (80)
Phentermine-topiramate	−9.4% (56 weeks)	Mean A1C change • −1.6% PHEN-TOP vs. −1.2% PBO at 56 weeks (<i>P</i> = 0.0381) (81) Achieved A1C <7% • 51% to 63% PHEN-TOP vs. 39% to 40% PBO at 56 weeks (81)
Naltrexone-bupropion	−5.0% (56 weeks)	Mean A1C change • −0.6% NB vs. −0.1% PBO at 56 weeks (82) Achieved A1C <7% • 44.1% NB vs. 26.3% PBO at 56 weeks (82)
Orlistat	−6.2% (52 weeks)	Mean A1C change • −0.3% ORL vs. +0.2% PBO at 52 weeks (83)
Adults with hypertension: blood pressure reduction (Recommendation 2.9)		
Tirzepatide	−15.0% to −20.9% (72 weeks)	Mean SBP change • −7.2 mmHg TZP and −1.0 mmHg PBO at 72 weeks (18) • PBO-adjusted ABPM change of −7.4 mmHg for TZP 5 mg, −10.6 mmHg for TZP 10 mg, and −8.0 mmHg for TZP 15 mg on ABPM at 36 weeks (95) Mean DBP change • −4.8 mmHg TZP and −0.8 mmHg PBO groups at 72 weeks (18)
Semaglutide	−14.9% (68 weeks)	Mean SBP change • −6.2 mmHg SEMA and −1.1 mmHg PBO at 68 weeks (<i>P</i> <0.001) (42) Mean DBP change • −2.8 mmHg SEMA and −0.4 mmHg PBO at 68 weeks (42)

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TABLE 2.5 Summary of key evidence to inform obesity medication selection, by populations with specific obesity-related diseases and complications* (Continued)

Obesity medication	Mean weight change with obesity medication* (time point)	Select outcomes relevant to population
Orlistat	−5.3% (208 weeks)	Mean SBP change • −7.3 mmHg ORL vs. −5.2 mmHg PBO at 52 weeks ($P < 0.01$); −4.9 mmHg ORL vs. −3.4 mmHg PBO at 208 weeks ($P < 0.01$) (69) Mean DBP change • −3.6 mmHg ORL vs. −2.6 mmHg PBO at 52 weeks ($P < 0.01$); −2.6 mmHg ORL vs. −1.9 mmHg PBO at 208 weeks ($P < 0.01$) (69)
Phentermine- topiramate	−7.8% to −9.89% (56 weeks)	Mean SBP change • −5.6 mmHg PHEN-TOP 15/92 mg vs. −2.4 mmHg PBO at 56 weeks ($P < 0.001$) (40) Mean DBP change • 3.8 mmHg PHEN-TOP 15/92 mg vs. −2.7 mmHg PBO at 56 weeks ($P = 0.003$) (40)
Liraglutide	−8.0% (56 weeks)	Mean SBP change • −4.2 mmHg LIRA and −1.5 mmHg PBO at 56 weeks ($P < 0.001$) (41) Mean DBP change • −2.6 mmHg LIRA and −1.9 mmHg PBO groups at 56 weeks ($P < 0.001$) (41)
Adults with obesity and established atherosclerotic cardiovascular disease: reduce cardiovascular events (Recommendation 2.10)		
Semaglutide	−9.4% (104 weeks)	• MACE HR 0.80 (95% CI 0.72–0.90; $P < 0.001$) SEMA vs. PBO over a mean follow-up of 40 months (4)
Liraglutide	−5.7% to −9.2% (≥32 weeks)	• MACE HR 0.42 (95% CI 0.17–1.08) LIRA vs. comparators ≥32 weeks (106) • MACE HR 0.87 (95% CI 0.78–0.97; $P < 0.001$) for liraglutide vs. PBO over a median follow-up of 3.8 years (5)
Tirzepatide	−11.9% to −17.8% (72 weeks)	• Change in 10-year ASCVD risk score −23.5% to −16.4% TZP vs. +12.7% PBO at 72 weeks ($P < 0.001$) (108)
Adults with obesity and heart failure with preserved ejection fraction: reduce heart failure symptoms or events (Recommendation 2.11)		
Semaglutide	−9.8% to −13.3% (52 weeks)	Heart failure symptoms • Mean KCCQ-CSS change 16.6 SEMA vs. 8.7 PBO at 52 weeks ($P < 0.001$) (47) • Mean KCCQ-CSS change 13.7 SEMA vs. 6.4 PBO at 52 weeks ($P < 0.001$) (255) • Mean KCCQ-CSS change 15.0 SEMA vs. 7.5 PBO at 52 weeks ($P < 0.0001$) (118) Heart failure events • SELECT trial ◦ Composite HF outcome HR 0.82 (95% CI 0.71–0.96) SEMA vs. PBO over mean follow-up of 40 months (4) • Pooled analysis (SELECT, FLOW, STEP-HFpEF, STEP-HFpEF DM) ◦ CV death or HF event HR 0.69 (95% CI 0.53–0.89; $P = 0.0045$) SEMA vs. PBO at ≥52 weeks (121) ◦ Worsening HF event HR 0.59 (95% CI 0.41–0.82; $P = 0.0019$) SEMA vs. PBO at ≥52 weeks (121)
Tirzepatide	−13.9% (52 weeks)	Heart failure symptoms • Mean KCCQ-CSS change 19.5 TZP vs. 12.7 PBO at 52 weeks ($P < 0.001$) (48) • Between-group median KCCQ-CSS difference 6.9 favoring TZP over PBO at 52 weeks ($P < 0.001$) (123) Heart failure events • CV death or worsening HF event (HF hospitalization, intravenous drugs in an urgent care setting, or intensification of oral diuretic therapy) HR 0.62 (95% CI 0.41–0.95; $P = 0.026$) TZP vs. PBO over a median follow up of 104 weeks (48)

Continued on p. 15 »

TABLE 2.5 Summary of key evidence to inform obesity medication selection, by populations with specific obesity-related diseases and complications* (Continued)

A1C, hemoglobin A1c; ABPM, ambulatory blood pressure monitoring; AHI, apnea-hypopnea index; ASCVD, atherosclerotic cardiovascular disease; CV, cardiovascular; DBP, diastolic blood pressure; HF, heart failure; HFpEF, heart failure with preserved ejection fraction; HR, hazard ratio; KCCQ-CSS, Kansas City Cardiomyopathy Questionnaire clinical summary score; KOOS, Knee injury and Osteoarthritis Outcome Score; LIRA, liraglutide; MACE, major adverse cardiovascular event (cardiovascular death, nonfatal myocardial infarction, nonfatal stroke); MASH, metabolic dysfunction-associated steatohepatitis; NB, naltrexone-bupropion; ORL, orlistat; PAP, positive airway pressure; PBO, placebo; PHEN-TOP, phentermine-topiramate; SBP, systolic blood pressure; SEMA, semaglutide; T2D, type 2 diabetes; TZP, tirzepatide; WOMAC, Western Ontario and McMaster Universities Osteoarthritis Index. *Of note, phentermine is not included in this table, as all outcomes reported require long-term trials and phentermine has only been studied in short-term RCTs (28 weeks or less) to date.

Obesity is a leading driver of heart failure with preserved ejection fraction (HFpEF) onset and progression (111–115), and individuals with obesity and HFpEF display unique clinicopathological features (e.g., increased

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related to the weight-reduction magnitude achieved with the obesity medication (139,140). To date, the greatest magnitude of benefit has been with tirzepatide. In the SURMOUNT-OSA trials, once-weekly tirzepatide (10 or 15 mg) significantly reduced AHI by more than 20 events/h compared with placebo in adults with obesity and moderate-to-severe OSA, among individuals treated with and without positive airway pressure (PAP) (6). Mean weight reduction with tirzepatide was $\geq 15\%$. Over 40% of SURMOUNT-OSA participants treated with tirzepatide experienced improvements in OSA severity to below typical thresholds to recommend PAP therapy. In short-term RCTs, phentermine-topiramate and liraglutide 3.0 mg reduced AHI by 15 and 6 events/h, respectively, compared with placebo among people with obesity and moderate-to-severe OSA (139,140). Inclusion of obesity medications may also have added benefits of improved systolic blood pressure, glycemia, and body weight reduction that individuals with OSA may be unlikely to achieve with PAP therapy alone (141,142). In addition to established OSA treatment approaches (e.g., PAP, airway stimulation, and splinting devices), the evidence supports obesity medications as part of the treatment plan for people with obesity and OSA.

Osteoarthritis

Osteoarthritis is an obesity-related disease, particularly knee osteoarthritis (143,144). Weight gains of $\geq 10\%$ are associated with worsening pain and physical function (145), and weight reduction of similar magnitude may result in symptom improvements (146). In adults with obesity and moderate osteoarthritis, the treatment plan should prioritize a GLP-1RA or dual GIP/GLP-1RA with potential to improve osteoarthritis symptoms (Fig. 2.2 and Table 2.5). In a 68-week trial involving individuals with obesity and moderate knee osteoarthritis, semaglutide 2.4 mg significantly improved knee pain scores and reduced body weight compared with placebo (49); participants also experienced greater improvements in physical function. Less evidence exists regarding the use of liraglutide in osteoarthritis, which may only have modest benefits in physical function (147). A retrospective cohort study found that individuals prescribed semaglutide, liraglutide, or tirzepatide had a reduced risk of osteoarthritis diagnosis (148). Individuals prescribed tirzepatide had a significantly lower risk of osteoarthritis compared with individuals prescribed semaglutide or liraglutide.

Long-term Management of Obesity Medications

Recommendations

2.15 Obesity medications requiring dose escalation should be initiated at a low dose and uptitrated gradually, as needed, based on tolerability **A** and clinical response **E**.

2.16 Continue obesity medications after reaching treatment goals to maintain health benefits, as discontinuation often results in weight recurrence and worsening or reemergence of cardiometabolic risk factors or obesity-related diseases

and complications **A**. The individualized maintenance dose should balance efficacy, health benefits, and tolerability and therefore may not necessarily be the maximum approved dose **A**.

2.17 In individuals with an inadequate response to current obesity medication, intensify treatment to improve outcomes. Dose escalation to maximum dose is recommended **A**. If maximum tolerable dose is already used, then consider switching to an alternative obesity medication **C**, adding intensive lifestyle therapy **B**, combining obesity medications **C**, or referring to metabolic-bariatric surgery **A**.

Gradual dose escalation is a critical strategy in obesity pharmacotherapy to optimize efficacy while improving tolerability. Many obesity medications have dose-dependent adverse effects, particularly gastrointestinal symptoms and neuropsychiatric effects (e.g., mood changes), that necessitate a stepwise escalation (39–42). For example, the SURMOUNT-1 RCT demonstrated that a 20-week dose-escalation period was essential to reaching the target therapeutic dose of tirzepatide while maintaining tolerability (18). Tirzepatide dose escalation often occurs more gradually in real-world clinical practice than what occurred in trials (149), reflecting the importance of adjustments based on individual response. Experiencing adverse effects has been significantly associated with obesity medication discontinuation (59). Individuals who experience early intolerance to high doses often benefit from remaining on low doses longer before advancing. Health care professionals may consider this approach to improve tolerability, particularly for older individuals, people with gastrointestinal sensitivities, or individuals with obesity-related diseases or complications.

As obesity is a chronic disease, long-term pharmacotherapy is typically needed after treatment goals are achieved to sustain benefits. Real-world evidence thus far shows that long-term use of obesity medications can maintain $\geq 10\%$ weight reduction (150). Health care professionals and stakeholders who determine benefits coverage for health insurance should be aware that clinical trials show that discontinuation of obesity medications results in recurrence of most of the weight lost within 1 year and reversal of prior cardiometabolic improvements (107,151–153). Recent research has identified obesity medication costs as a factor contributing to individuals' discontinuing treatment (154). Loss of insurance coverage for obesity medications makes these treatments unaffordable for many individuals and contributes to their feelings of hopelessness, anger, and stigmatization (155). Obesity medications should be continued after reaching treatment and weight reduction goals to maintain health benefits and avoid weight recurrence (107,150,151,153,156).

The long-term maintenance dose of obesity medication should be tailored to optimize health benefits while minimizing adverse effects; multiple clinical trials have employed a “maximum tolerated dose” strategy to improve

RCT found that naltrexone-bupropion with lifestyle counseling resulted in 6.1% weight reduction at 1 year (39) as compared with another RCT that found naltrexone-bupropion with IBT resulted in 9.3% weight reduction at 1 year (22). Similarly, an RCT found that tirzepatide with lifestyle counseling resulted in 20.9% weight reduction (18), while another RCT found that combining tirzepatide with IBT resulted in a 24.5% weight reduction (25). These results may suggest that combining obesity medication with IBT may improve outcomes relative to obesity medication with counseling alone. Referral should be considered where there is access to such programs.

For individuals who have a partial treatment response to the current obesity medication, it may be reasonable to combine this medication with another obesity medication. Health care professionals can target different weight-regulating pathways by combining obesity medications; in fact, weight reduction achieved by low-dose combination therapy may exceed that of high-dose monotherapy. For example, an RCT found that weight reduction achieved with phentermine-topiramate 7.5/46 mg was greater than phentermine 15 mg alone (10.4% vs. 7.4%, respectively) (50). Among individuals with T2D, the addition of naltrexone-bupropion to a GLP-1RA resulted in 5.2% greater weight reduction than that for individuals on placebo and GLP-1RA (168). However, a small pilot study found that the addition of phentermine to liraglutide did not produce additional clinically meaningful weight reduction in individuals who had already lost 12.6% of initial weight with liraglutide alone (169). The use of multiple obesity medications for chronic weight management mirrors treatment paradigms established for other chronic diseases, and this approach is also supported by long-term data in retrospective cohorts (150,156). However, there are no RCTs examining how to best escalate therapy for individuals with inadequate treatment response.

Metabolic-bariatric surgery should be offered to eligible individuals with inadequate treatment response to lifestyle modification and pharmacotherapy. Metabolic-bariatric surgery produces substantial weight reduction and achieves multiple health benefits including reduced blood pressure (170), improved glycemia among people with T2D (171), and reduced cardiovascular events (172). In addition, several studies have shown the benefit and safety of adding obesity medication after an individual has undergone metabolic-bariatric surgery (173–175). Therefore, obesity medications may also be added to augment weight reduction and health benefits in individuals who have undergone surgery where a weight plateau occurs before achieving their treatment goals or who are experiencing weight recurrence. Health care professionals should be aware of nonsurgical procedures to treat obesity, and a future section of the “Standards of Care for Overweight and Obesity” will provide recommendations regarding surgical and nonsurgical procedures to treat obesity.

18

Recommendations

2.19 Counsel and regularly monitor individuals using obesity medications to ensure adequate nutritional intake, with particular attention to preventing protein insufficiency and micronutrient deficiencies during the active phase of weight reduction **E**

When making nutrition recommendations, health care professionals should consider an individual's medical history, dietary intake and nutritional status, social determinants of health, personal values, and preferences. A meta-analysis concluded that personalized nutrition interventions are more effective at improving dietary intake than usual care, with mixed improvements in weight and waist circumference (16 studies) (176); however, another meta-analysis found that participant choice in dietary intervention did not confer a weight reduction benefit (9 studies) (177). Given that in-depth nutritional counseling may be challenging in clinical practice (178–180), referral to a registered dietitian nutritionist (RDN) for nutritional guidance should be considered for people prescribed obesity medications. RDNs use medical nutrition therapy—a counseling process that includes nutritional assessment, diagnosis, intervention, and monitoring and typically results in the prevention, delay, or management of disease (181). RDN-delivered obesity interventions improve BMI, waist circumference, blood pressure, and quality of life (182). In a systematic review (26 RCTs), RDN consultations were effective for improving nutritional quality as well as weight reduction and diabetes outcomes in the primary care setting (183). RDNs may also address other important behaviors including physical activity, sleep hygiene, stress management, problem solving, and goal setting (184).

In general, adults with obesity have lower intakes of vitamins A, C, D, and E as well as calcium and magnesium compared with adults without obesity (187). The appetite suppressive effects of obesity medications lead to caloric reduction (188,189), and this reduced food consumption may be insufficient to meet essential micronutrient requirements in some individuals (190,191). Multivitamin mineral supplements may be considered for individuals who consume less than 1,200 kcal/day, exclude micronutrient-rich food groups from their usual intake (e.g., fruits and vegetables, whole grains, proteins, and nuts and seeds), are strict vegetarians, have underlying health conditions that impair nutrient absorption, are older (aged >50 years), or experience excessive weight reduction (192). Multivitamin mineral supplement use has been associated with lower risk of nutrient deficiencies (187,193). A daily multivitamin supplement is also recommended with orlistat therapy due to decreased absorption of fat-soluble vitamins (194). If nutritional intake remains inadequate due to obesity medication effects, the health care professional should consider dose reduction of the obesity medication.

All weight-reduction interventions are associated with reductions in fat mass and lean body mass (LBM), although the proportions of the two components vary over time and are determined by multiple factors (e.g., protein intake, exercise, sex, age, and baseline adiposity). In general, many nutrition intervention studies report that fat mass loss contributes to ~75% of body weight reduction, while LBM contributes to ~25% of body weight reduction (195). A concern over the risk of excessive skeletal muscle loss, a component of LBM, has been raised with the use of obesity medications, particularly medications that result in $\geq 15\%$ weight reduction. In meta-analyses of GLP-1RA obesity medications, LBM loss accounts for 25% to 30% of total weight reduction (196,197). Similarly, a substudy of participants on tirzepatide found that LBM loss accounts for 25% of total weight reduction (198). Despite lean mass loss, questionnaires consistently demonstrate improvements in physical function and quality of life with weight reduction (199,200). However, older adults with obesity or adults with severe obesity-related diseases or complications may be vulnerable to later onset of sarcopenic obesity—a condition characterized by reduced skeletal muscle mass and function (201)—and lower bone mineral density due to excessive weight reduction. Health care professionals should consider regularly assessing muscle strength and function by one or more office-based techniques (e.g., chair-to-stand, handgrip, knee-extensor strength, and walk test) (201) and using functional instruments (e.g., SARC-F questionnaire) (202).

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Special Populations and Circumstances

2.24b If an obesity medication is unavailable (e.g., in short-age), a switch to a different FDA-approved obesity medication is recommended, as clinically appropriate **E**. Upon resolution of the unavailability, reassess the appropriateness of resuming the original FDA-approved obesity medication **E**.

2.25 Clinical practices that are unable to implement the recommended infrastructure to support the pharmacologic treatment of obesity should consider referring people to

2.30 Clinical practices should consider allocating resources to manage prior authorizations for obesity medications as a strategy to reduce the likelihood that these requirements alter health care professionals' treatment decisions **E**.

Given that education and training on obesity treatment is lacking or insufficient in many medical schools, residency programs, and fellowships (235), many health care professionals are unfamiliar with obesity medications and have limited experience with their prescribing (236,237).

Prior to prescribing an obesity medication, health care professionals should complete a comprehensive evaluation that includes eliciting a history relevant to obesity, performing a physical examination, and obtaining laboratory testing to evaluate for obesity-related diseases and complications that may inform selection of obesity medication. Table 2.6 provides a brief overview of select history and physical exam elements that should be obtained. A future section of the “Standards of Care for Overweight and Obesity” will provide recommendations and additional details regarding the diagnosis, evaluation, and staging of obesity. A comprehensive physical examination should be performed to evaluate for causes of obesity (37). Height and weight should be measured, and BMI should be calculated to categorize the class of obesity. Waist circumference should be measured with the proper technique, and waist-to-height ratio should be calculated (≥ 0.5 indicates central adiposity with increased risk of T2D and cardiovascular disease) (240,241). Blood pressure should be measured with an appropriately sized cuff to minimize measurement error; importantly, blood pressure values are used to inform obesity medication selection (e.g., elevated blood pressure is a contraindication to naltrexone-bupropion and phentermine).

Health care professionals should document whether an individual has a history of the following obesity-related diseases and complications that may inform obesity medication selection: prediabetes, T2D, hypertension, ASCVD, HFpEF, MASH, OSA, or osteoarthritis. If the individual has not been evaluated for these conditions, the health care professional should obtain a history and complete a physical examination pertinent to the condition and obtain additional testing, as appropriate, to establish a diagnosis (Table 2.6). Clinical practices should be prepared to conduct comprehensive initial evaluations and to continue monitoring identified obesity-related diseases and complications over time.

As obesity medications must be used in combination with lifestyle modification, clinical practices should determine their approach to delivering this care, either as a service provided within the practice or in collaboration with health care professionals or services outside the practice. Lifestyle interventions can be delivered within the primary care setting. A systematic review (56 RCTs) showed that primary care-based interventions for adults with obesity were effective in reducing weight and waist circumference (242); the average intervention duration

History elements	Physical examination elements	Diagnostic testing elements
Weight history • Weight gain pattern and body weight trajectory • Highest and lowest body weight as an adult • Characteristics at the onset of overweight (e.g., age, life events, contributing factors) • Past or current use of weight-promoting medications • Past and current treatments and responses, including lifestyle, medication, procedures, and surgery • Past and current disordered eating • Personal goals and readiness to change Lifestyle behaviors • Eating patterns • Physical activity • Stress and coping • Sleep • Personal and cultural preferences for nutrition and physical activity Past medical history of obesity-related diseases and complications† • Prediabetes • Type 2 diabetes • Hypertension • Atherosclerotic cardiovascular disease • Heart failure with preserved ejection fraction • Metabolic dysfunction–associated steatohepatitis or metabolic dysfunction–associated steatotic liver disease • Obstructive sleep apnea • Knee osteoarthritis • Others People of childbearing potential • Current contraception strategy • Upcoming plans for pregnancy	Anthropometrics • Weight and height (measured) • Calculate BMI • Waist circumference (measured), as appropriate‡ • Calculate waist-to-height ratio, as appropriate‡ Vital signs • Blood pressure (measured) • Heart rate (measured) Physical examination • Cardiac exam • Pulmonary exam	Laboratory testing • A1C • Electrolytes including Cr and eGFR • AST, ALT, and platelet count to calculate FIB-4 • TSH • Lipid panel • Urine pregnancy test Other testing, as clinically appropriate • Echocardiogram • Vibration-controlled transient elastography • Sleep study • Knee X-ray

was approximately 12 months. Another systematic review (12 RCTs) in adults with obesity receiving behavioral treatment through primary care found that interventions prescribing reduced energy intake (e.g., ≥ 500 kcal/day deficit) and increased physical activity with traditional behavioral therapy were associated with greater weight reduction than interventions without these three specific components (243). In addition, more treatment sessions delivered by trained interventionists (in-person or remote) were associated with greater weight reduction. As appropriate, clinical practices may consider referring individuals with obesity to IBT or structured lifestyle programs in their community. The effectiveness of combining obesity medication with a structured lifestyle program has also been demonstrated with an employer-based weight management program (165) and within the Veterans Health Administration's MOVE! weight

Following up with individuals started on obesity medications is recommended at least monthly for the first 3 months of treatment to support medication-taking behavior by assessing efficacy, managing adverse effects, and monitoring health status. Adverse effects from obesity medications are more likely to occur at initiation and with dose escalations (60,61), and frequent contact with the health care team may help support individuals

experiencing these effects. Table 2.1 includes strategies on how to manage common adverse effects for each obesity medication; health care professionals should also develop a differential diagnosis for significant adverse effects and exclude other potential causes or contributors to the symptoms. Unless clinical circumstances (such as poor medication tolerability) or other considerations (such as financial expense) suggest otherwise, individuals who achieve sufficient early weight reduction with an obesity medication (typically defined as $\geq 5\%$ weight reduction after 3 months) should continue the medication long term. Modeling from clinical trials shows that early responders have improved long-term outcomes (244,245). When early weight reduction results are modest, the benefits of ongoing treatment need to be examined in the context of dose-escalation progress, treatment tolerance, improvements in obesity-related diseases and complications, the availability of other treatment options, and overall treatment burden. During this initial period, follow-up can be delivered via telehealth with interprofessional team members to facilitate timely and effective care. An RCT evaluating the comparative effectiveness of a behavioral weight-reduction intervention found that both in-person and remotely delivered interventions led to significant weight reduction compared with control and that the weight change did not differ significantly between the two intervention groups (246). Real-world evidence also supports the use of telehealth care delivered by clinical practices, which achieves similar outcomes to in-person visits (247–249).

After this initial 3-month period, regular follow-up care with individuals on obesity medications is needed to continue monitoring the efficacy and safety of the treatment plan. Real-world data from an academic obesity medicine clinic found that a greater number of follow-up visits was associated with greater weight reduction (150), which underscores the importance of regular interaction between individuals with obesity and health care professionals. Health care professionals should arrange for follow-up at least every 3 months for the remainder of the first year of treatment and plan for follow-up at least every 6 months for long-term management. Ongoing monitoring of the achievement and maintenance of treatment and weight-reduction goals is recommended. For individuals not reaching or maintaining treatment goals, treatment inertia should be avoided by reevaluating ongoing obesity therapies and intensifying treatment with additional approaches. Shared decision-making should be used to determine the best long-term obesity treatment approach. See the section above on Long-term Management of Obesity Medications for additional information.

Health care teams should be knowledgeable about health insurance formularies and obesity medication coverage requirements, as well as eligibility for medication assistance programs, copayment reduction cards, and direct-

to-consumer cash pay options. Clinical practices should anticipate that most prescriptions for obesity medications will require prior authorization. U.S. health care professionals have reported that prior authorization alters their prescribing behaviors generally (250). A systematic review (46 studies) found that prior authorization and step therapy requirements were associated with lower medication use, which negatively affected medication-taking behaviors and clinical outcomes (251). Therefore, clinical practices should consider allocating resources to manage prior authorizations for obesity medications. Pharmacists or pharmacy technicians may also be able to assist with this process. Given that insurance coverage and costs may influence treatment decisions, payors should cover evidence-based obesity medications, particularly for people with obesity-related diseases and complications, to improve access and outcomes. Real-world research found that individuals who combined obesity medication with an intensive lifestyle program were significantly more likely to achieve clinically significant weight reductions and improve cardiometabolic markers (e.g., blood pressure and A1C) at 6, 12, and 24 months as compared with an intensive lifestyle program alone (166). Notably, the individuals prescribed obesity medication with the program also had fewer inpatient hospitalizations and emergency department visits than those who completed the lifestyle program alone, which resulted in lower medical costs.

Conclusions

Stakeholders, including individuals with obesity, health care professionals, policy makers, insurers, and employers, increasingly recognize the need for safe and effective pharmacotherapy to treat obesity. The ADA's Obesity Association encourages health care professionals to adopt and implement the present recommendations on obesity medications as part of a comprehensive obesity treatment plan for adults. Evidence supports the use of obesity medications in combination with lifestyle interventions to achieve long-term health and weight reduction goals. Obesity medications should be discussed with all individuals with obesity, and treatment plans for individuals with obesity who have or are at high risk of obesity-related diseases and complications should prioritize use of obesity medications with demonstrated benefit in improving the individuals' obesity-related diseases. Given the various pharmacologic approaches now in development, additional obesity medications are likely to come to market in coming years that may further expand treatment options.

Appendix

American Diabetes Association Professional Practice Committee for Obesity. Committee members include Kimberly A. Gudzone,^{||***} Caroline M. Apovian, Vanita R. Aroda, Louis J. Aronne,^{***} Kirthikaa Balapattabi,^{**} Allison K.

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